Pickleball Matched Pairs Worksheet

Pickleball is a fast-paced sport blending tennis, badminton, and ping-pong, played with paddles and a plastic ball on a small court. College tournaments feature singles, doubles, and mixed doubles in a best-of-three format, emphasizing strategy, teamwork, and precision. Player performance is often tracked using the DUPR (Dynamic Universal Pickleball Rating) system, which rates skill from 2.000 to 8.000 based on match results, opponent strength, and recent play. A Reliability Score (1–100%) reflects how accurate a rating is, with 60%+ considered dependable.

Read in the Dataset:

1 - Creating a boxplot

A) Create a boxplot comparing Singles and Doubles Reliability scores.

B) What do you notice about the center, spread, and any outliers in the distributions? Which category appears to have a higher average reliability?

We want to perform a hypothesis test to test if there is a significant difference between singles and doubles reliability. Because each player was measured twice (single and double reliability), we want to use a matched pairs t-test to perform the hypothesis test.

2 - Checking assumptions

A) State and evaluate the three assumptions needed for a paired t-test. You may assume independence. Use R code to help justify normality and check for outliers.

1. Independence:
2. Normality:
3. No Extreme Outliers:

3 - Performing a matched pairs hypothesis test manually.

Let be the true mean difference (Doubles - Singles) - because doubles reliability mean is higher than singles reliability.

A) State the null and alternative hypothesis.

B) Create a new column labeled `Difference` that takes Doubles Reliability minus Singles Reliability.

C) Calculate the mean difference of the `Difference` column. Report your result.

D) Calculate the standard deviation of the `Difference` column. Report your result.

E) Calculate the t-statistic. The formula is below in which you can do it by hand or do it using r. Report your result.

$$t = \frac{\bar{D}}{s_D/\sqrt{n}}$$

F) Calculate the degrees of freedom. Report your result.

G) Calculate the p-value for two-tailed hypothesis. To do this do in R code - 2 * pt(-abs(t-statistic), df = degrees of freedom). Report your result.

H) Use the p-value to make a conclusion.

We can also do this process using the t.test function in r which allows us to perform the hypothesis test quicker.

4 - Performing a matched pairs hypothesis test using R function:

A) perform a matched pairs hypothesis t-test. Include all parts of a formal hypothesis test.

B) Interpret the confidence interval results given in the output in part A.

You could also interpret it this way… On average, doubles reliability is between 12.4% and 16.7% higher than singles reliability.

A possible explanation for this is that doubles matches are more commonly played than single matches.

If you notice you gained two different p-values for both manually and using R. This is because the t.test() function in R does not report p-values smaller than 2.2e-16. When calculated manually, however, the full precision of the test statistic can be used to generate an even smaller p-value. This does not change the outcome of the results. Both methods will result in the same statistical conclusion.